

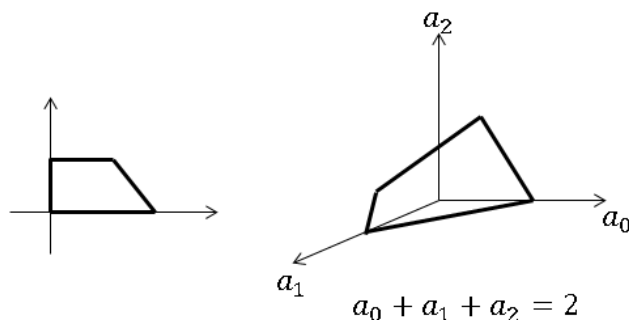
Lecture #6

- (0) Jose scribing
- (1) Homogeneity
- (2) Boolean operations, closure and projection

Last time, we saw the definition of $\mathbb{P}_k^n$ and claim that projective space could simplify some geometric statements, e.g: vanishing resultants.

Definition: The degree d homogenization of a polynomial $f \in \mathbb{C}[x_1, \dots, x_n]$ (with degree $f \leq d$) is just $\tilde{f} = x_0^d f(x_1/x_0, \dots, x_n/x_0)$.

Example: The (degree 2) homogenization of $x_1^2 - x_1x_2 + 3 - 7x_2$ is just $x_1^2 - x_1x_2 + 3x_0^2 - 7x_0x_2$.



Proposition: $\text{Newt}(\text{any homogenization}) \subset \text{hyperplane of the form } a_0 + \dots + a_n = \text{const.}$

Proposition: f has a root at $(\zeta_1, \dots, \zeta_n) \in \mathbb{C}^n \Leftrightarrow \tilde{f}$ has a root at $[1 : \zeta_1 : \dots : \zeta_n] \in \mathbb{P}_{\mathbb{C}}^n$.

Note: The notion of "roots" thus make sense in $\mathbb{P}_{\mathbb{C}}^n$.

Note: Roots of the form $[0 : \zeta_1 : \dots : \zeta_n]$ are called roots at ∞ .

Corollary (Variant of FTA): A homogeneous (not identically zero) $f \in \mathbb{C}[x_0, x_1]$ factors as $f(x_0, x_1) = \prod_{i=1}^d (\alpha_i x_0 + \beta_i x_1)$ for some $[\alpha_i, \beta_i] \in \mathbb{P}_{\mathbb{C}}^1$.

Note: roots of f are: $[x_0 : x_1] \in \{[\beta_i : -\alpha_i]\}$

Theorem (on vanishing resultants):

If $f(x_1) = a_0 + \dots + a_d x_1^d$ and $g(x_1) = b_0 + b_1 x_1 + \dots + b_e x_1^e$ then $\tilde{f} = \tilde{g} = 0$ has a root in $\mathbb{P}_{\mathbb{C}}^1 \Leftrightarrow \text{Res}_{(d,e)}(f, g) = 0$. ($\tilde{f}$ deg d homog, $\tilde{g}$ deg e homog).

$\mathbb{P}_{\mathbb{C}}^1$ is also very useful because it is compact as metric space.

Example:

$\epsilon x^2 - 2\epsilon x + 1 = (\epsilon x - 1)^2$ has roots $\frac{1}{\epsilon}$ (mult 2).

So as $\epsilon \rightarrow 0$ this quadratic polynomial ceases being quadratic and the root has no well-defined limit.

But, if you homogenize (to degree 2) you get $\tilde{f} = \epsilon x^2 - 2\epsilon x_0 x_1 + x_0^2 = (\epsilon x_1 - x_0)^2 \Rightarrow$ root is $[\epsilon : 1]$.

So as $\epsilon \rightarrow 0$, you do have a well-defined limit!

Metric?

Definition: $\text{dist}([x_0, \dots, x_n], [y_0, \dots, y_n]) = \arccos\left(\frac{|x_0 y_0 + \dots + x_n y_n|}{\sqrt{x_1^2 + \dots + x_n^2} \sqrt{y_1^2 + \dots + y_n^2}}\right)$

We are now, at least, getting to some deeper topological (and metric) issues...

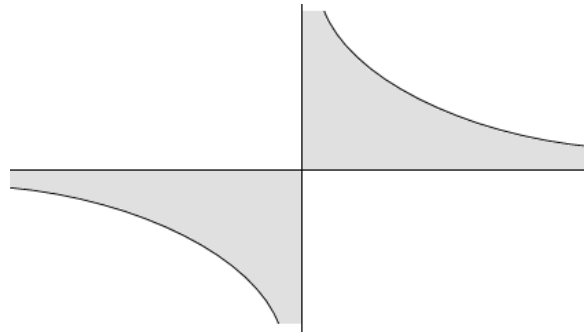
Definition: We call a subset $S \subset \mathbb{C}^n$ constructible $\Leftrightarrow$ it is a finite boolean combination of algebraic subsets of $\mathbb{C}^n$. i.e. finitely many application of $\cup, \cap$ and complement $(\cdot)^c$.

Definition: $S \subseteq \mathbb{C}^n$ is called algebraic $\Leftrightarrow$ it is the set of roots of polynomial system.

Projections: are subtle...

- (I) Algebraic sets are not closed under projections! E.g: Let $\pi : \mathbb{C}^2 \rightarrow \mathbb{C}$ be the map forgetting the second coordinate, i.e. $\pi(x, y) = x$ then $Z = \{(x, y) \in \mathbb{C}^2 | xy = 1\}$ is an algebraic set, but $\pi(Z)$ ain't!

It's easy to show that any algebraic subset of $\mathbb{C}$ is: $\emptyset$, $\{\text{finite}\}$, or $\mathbb{C}$. But $\pi(Z) = \mathbb{C} \setminus \{0\}$.



Note: $\pi(Z)$ is constructible

- (II) Chevalley's theorem on constructible sets (simplified):

If $\mathbb{C}^n \rightarrow \mathbb{C}^k$ is any coordinate projection then $\pi(\text{constructible}) = \text{constructible}$

- (III) Fundamental theorem of elimination theory (simplified):

Algebraic subsets of $\mathbb{P}_{\mathbb{C}^n}$ are closed under projection.

Example: Homogenize $x_1x_2 - 1 : x_1x_2 - x_0^2$.

Now, let $\tilde{Z} = \{[x_0 : x_1 : x_2] | x_1x_2 = x_0^2\}$

π (from before) does not extend trivially to all of $\mathbb{P}_{\mathbb{C}}^2$. E.g: where does $[0 : 0 : 1]$ go to?

However, you can extend it to all of the curve $\tilde{Z}$, and then get an onto projection.

More on this later...

We'll prove Chevalley's theorem in HW#3... the proof is based on resultants.

Corollary A: You can reduce solving $n \times n$ systems to solving univariate polynomials.

Corollary B: We can extend the notion of discriminant to higher dimensions.

Example (Corollary A):

$$\begin{aligned}x^4 - 3xy + y^3 &= 0 \\y^3 - 3x^2 + 1 &= 0\end{aligned}$$

How to solve? Elimination/projection...

So try this: think of these polynomials as living in $(\mathbb{C}[x])[y]$, then use resultants to eliminate y ...

$$\begin{aligned} & \text{Res}_{(3,3)}(x^4 - 3xy^2 + y^3, 1 - 3x^2 + y^3) \\ \det \left(\begin{bmatrix} x^4 & 0 & -3x & 1 & 0 & 0 \\ 0 & x^4 & 0 & -3x & 1 & 0 \\ 0 & 0 & x^4 & 0 & -3x & 1 \\ 1 - 3x^2 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 - 3x^2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 - 3x^2 & 0 & 0 & 1 \end{bmatrix} \right) &= \\ & -x^{12} - 9x^{10} - 24x^8 + 243x^7 - 9x^6 \\ & -162x^5 + 24x^4 + 27x^3 - 9x^2 + 1 \end{aligned}$$

This is a start: The roots of our last polynomial contain (but might not be exactly) the x -coordinates of the roots of our original system... but with a little more work, we can solve completely.

To illustrate Corollary B, we'll go more deeply, into A-discriminants

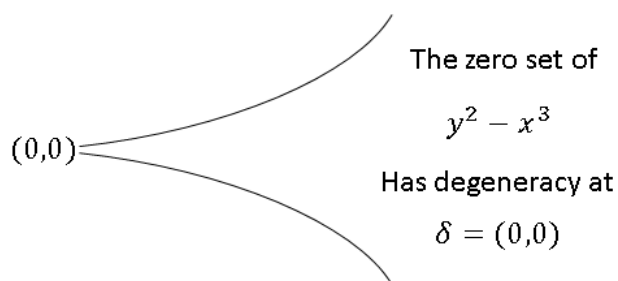
Definition: Given any finite $A = \{a_1, \dots, a_T\}$ (cardinality T) $\subset \mathbb{Z}^n$, we define $\nabla_A \subset \mathbb{P}_{\mathbb{C}}^{T-1}$ to be $\{[c_1, \dots, c_T] | c_1x^{a_1} + \dots + c_Tx^{a_T} \text{ has a degenerate root in } (\mathbb{C}^*)^n\}$

∇_A is the A-discriminant variety

Degenerate root (in $(\mathbb{C}^*)^n$)?

$A = \zeta = (\zeta_1, \dots, \zeta_n) \in (\mathbb{C}^*)^n$ with $f(\zeta) = 0$ ($\frac{\partial f}{\partial x_1}|_{\zeta} = \dots = \frac{\partial f}{\partial x_n}|_{\zeta} = 0$)

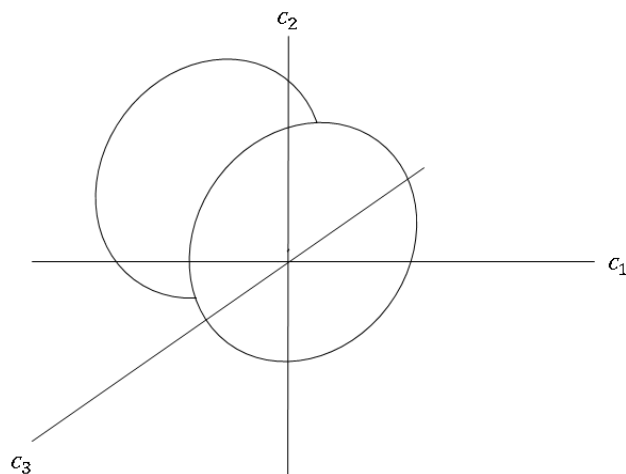
E.g:



Example: $A = \{0, 1, 2\}$ when does $c_1 + c_2x_1 + c_3x_1^2$ have a degenerate root? Exactly when:

$$c_3 \neq 0 \text{ and } c_2^2 - 4c_1c_3 = 0$$

So the closure is $\{[c_1 : c_2 : c_3] | c_2^2 - 4c_1c_3 = 0\}$. i.e. the $\mathbb{C}$ zero set in $\mathbb{P}_{\mathbb{C}}^2$ of $c_2^2 - 4c_1c_3$



... a double cone $\setminus \{c_3 = 0\}$